Identification of an Unknown Sample A

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November 27, 2023

Chem 2081-004

INTRODUCTION

The objective of this lab was to identify an unknown molecule by utilizing data from Mass Spectroscopy, Infrared, and NMR graphs. Each graph provided its own distinct form of information about the chemical's properties that, in combination, helped determine the unknown molecule. The Nuclear Magnetic Resonance graph was used to determine the number of hydrogen atoms in the molecule as well as how they were distributed throughout the structure. This type of graph uses individual magnetic resonance fields to make peaks along the graph showing the nature of the protons within the molecule. The Infrared Radiation Graph then indicated the types of functional groups present within the molecule by using infrared radiation to excite the bonds within the molecule causing them to absorb radiation. In addition to those, the Mass Spectroscopy graph was used to determine precise ions present in the molecule as well as its molecular mass. This skill enables scientists to identify unknown compounds through testing. It is essential for every scientist to be able to interpret each of these graphs after testing in order to validate the identity of compounds being worked with.

THEORY

By measuring the many kinds of bonds that exist between molecules, infrared spectroscopy generates a spectrum depending on how the molecules and bonds absorb the infrared radiation. Certain bonds in molecules become stimulated when they take in infrared radiation causing them to vibrate. The amplitude of these vibrations increase and become more excited if the radiation's frequency coincides with the bond's radiation. Differently shaped molecules produce various patterns of absorption because they do not absorb the same infrared energy in the same places. The functional group region and the fingerprint region are the two main regions of the spectrum. While some functional groups fall in the fingerprint region from 1200 $cm{-}1$ to 600 $cm{-}1$, most functional groups fall in the functional region

from 3600 $cm{-}1$ to 1200 $cm{-}1$. As a result of their varying radiation absorption capacities, single, double, and triple bonds appear at various locations on the graph. The range of single bonds that are formed between carbon, hydrogen, oxygen, and nitrogen is 3600 $cm{-}1$ to 2850 $cm{-}1$, so when peaks appear in this area of the graph, it is most likely a single bond that is stimulated and absorbing IR. Since the range of triple bonds between carbon and nitrogen is 2300 $cm{-}1$ to 2100 $cm{-}1$, most alkynes will be found in this region. Alkenes may be recognized in the region of 1850 $cm{-}1$ to 1620 $cm{-}1$, as this is where most double bonds containing carbon, oxygen, and nitrogen are located. An instrument known as an IR spectrometer is used to excite bonds and absorb frequencies. After the sample is put within the apparatus, the sample is exposed to IR light. A detector detects the amount of light absorbed by the sample after it passes through the sample. Next, an infrared plot displaying the light frequencies absorbed by the sample is created using this data. Every signal on the absorption spectrum has a corresponding structure or group on the molecule based on its intensity and position. By interpreting the graph, it can be concluded which functional groups are in the molecule.

Using mass spectroscopy, particles and molecules are analyzed based on their molecular weight and mass. Mass spectroscopy operates in four steps. Using a process known as electron impact ionization, the organic molecule is first transformed into positive ions inside the machine by losing one or more electrons. A high energy electron beam is used to bombard the molecule once it is introduced into the machine in the gas phase. This intermediate, known as a molecular ion (M)+, is both a radical and a high energy cation. The cations are then all accelerated to have the same kinetic energy in the second stage. A magnetic field deflects each cation in accordance with its mass during the third stage. Ions will deflect more if they are lighter in weight and more charged. The ion will be strongly positively charged and will be deflected if many electrons are lost in the first stage. The detector will provide the precise ion size and signal intensity. An analysis of a mass spectrum reveals multiple significant peaks.

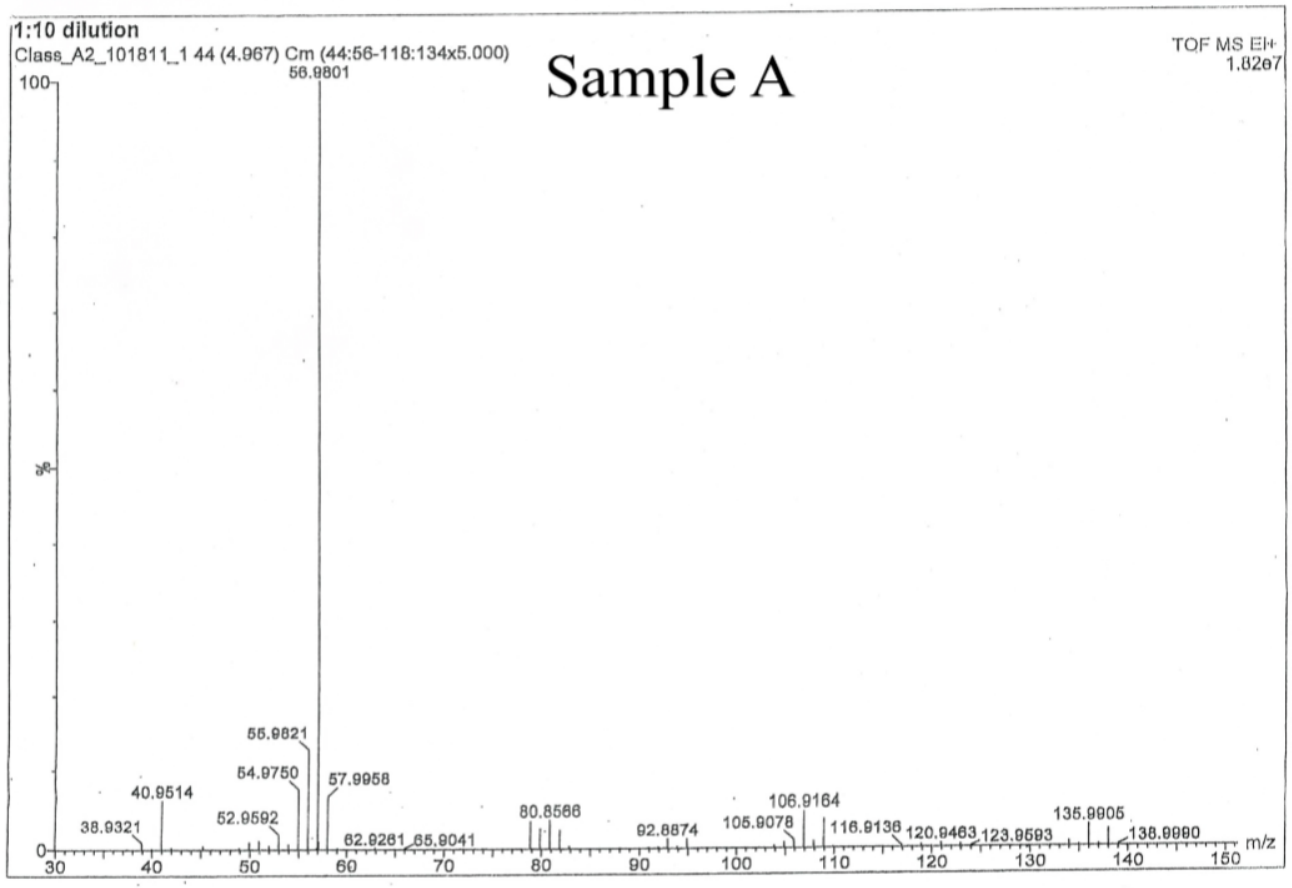
The base peak is the first. At 100% relative abundance, the base peak represents the most prevalent fragment ion. The variations in the peaks point to additional features of the molecule. The elements that make up the organic molecule can be identified using these differences. Elements with a large number of isotopes will also show multiple isotope peaks on their spectrum.

It is possible to determine the precise configuration and connectivity of atoms within a molecule using nuclear magnetic resonance. The location of particular functional groups and hydrogens in a molecule can be determined using NMR. Understanding the molecule's carbon and hydrogen framework is crucial for determining its identity and specific locations on the molecule. Nuclei and their reactions are studied by NMR using electromagnetic radiation. In opposition to the magnetic field, the nucleus flips to a higher energy state when a radio frequency is applied to it. Protons are said to be in resonance when there is a negligible energy difference from beginning to end. These waves are low radio frequency waves. A spectrum can be created once the sample has been subjected to the NMR machine in order to ascertain the protons' reactions and relative positions within the molecule. There are four main things to look for and interpret when examining an NMR spectrum. The quantity of signals on the spectrum should be examined first. The specific signal provides information about the surroundings of a single proton and how it interacts with the rest of the molecule. A single proton or several protons sharing the same signal due to their shared electronic environment could be present at that signal. The signal's location is the second thing to check. The difference between the proton's absorption frequency and the spectrometer's standard absorption is the signal's location, which is known as a chemical shift. The magnitude of the chemical shift is dependent on the location of the proton within the molecule. Either shielded or unshielded proton states are possible. A shielded proton will have a signal that is closer to zero on the right side of the spectrum. A proton's signal will be closer to ten on the left side of the spectrum if it is deshielded. The region beneath the signal is the third thing to search for. The

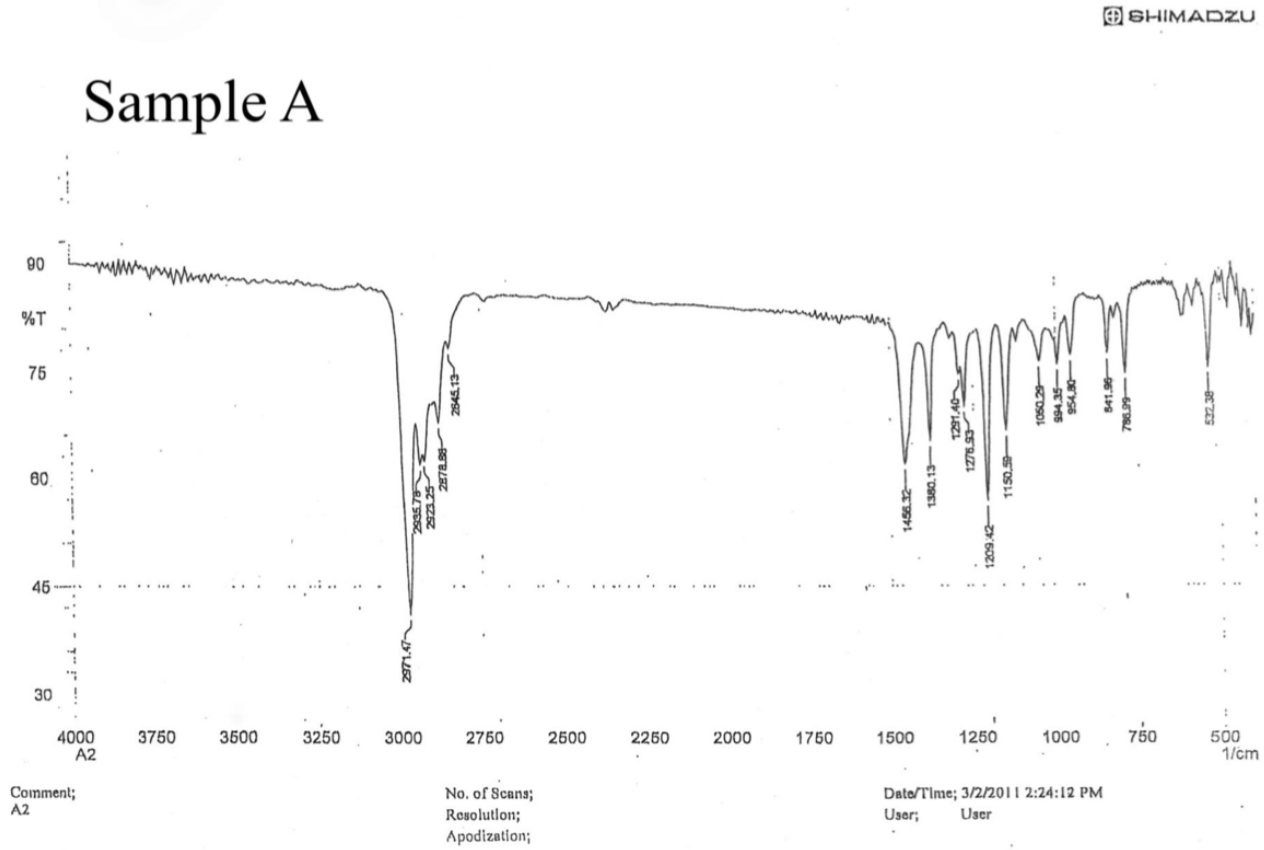
number of protons contained in the signal can be calculated using integration and used to find the area under the signal. Finally, the number of peaks at each individual signal is the fourth thing to look for on an NMR spectrum. We refer to this concept as multiplicity. Multiplicity is determined by the formula n+1 and is caused by the number of protons on the nearby carbon atom. The carbon has a multiplicity of four if its neighboring carbon has three protons. Determining the relationship between carbons and how they are all connected within the molecule is made easier by multiplicity.

DATA

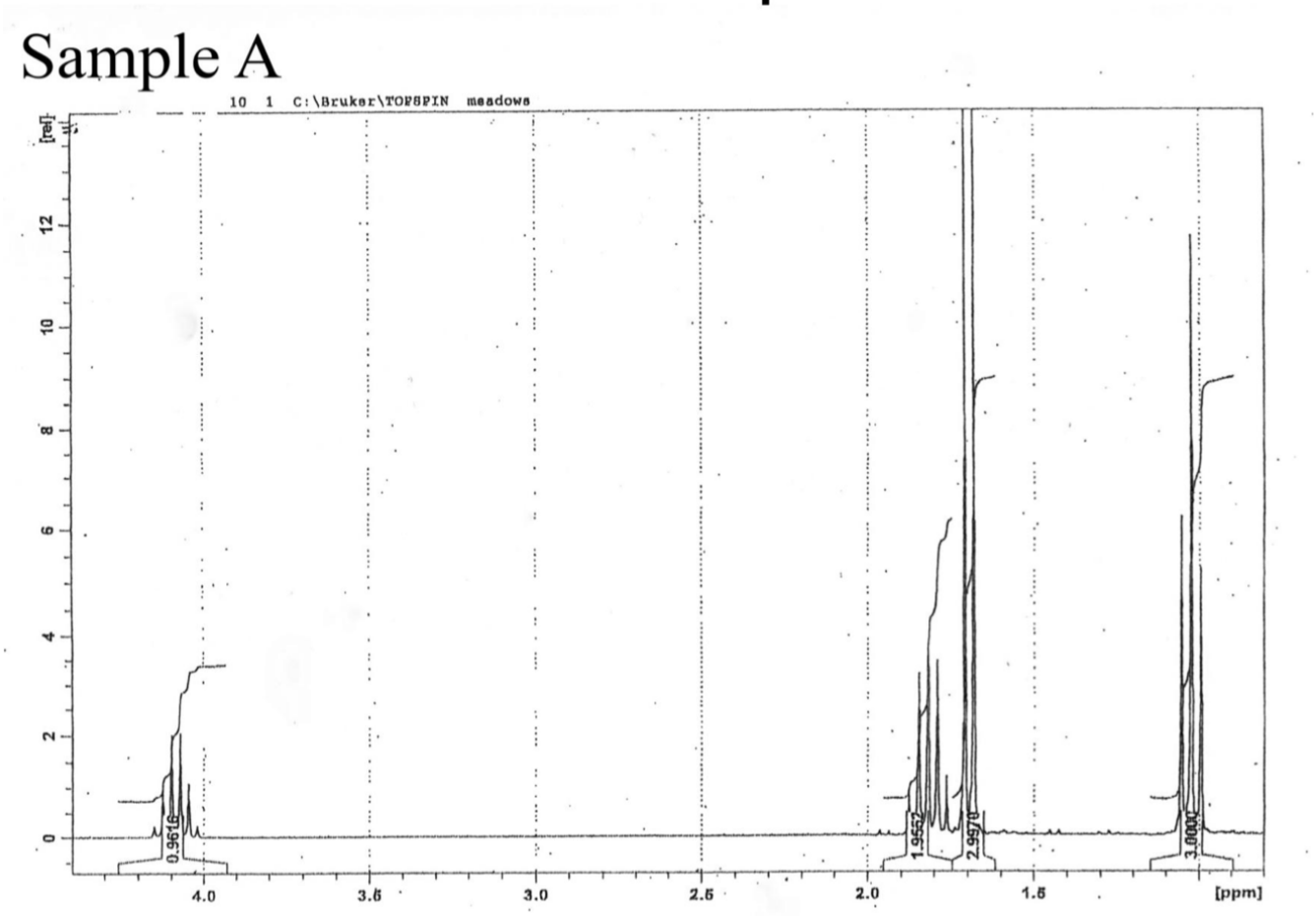
GCMS Sample A



FTIR Sample A



NMR Sample A



ANALYSIS

GCMS-

The base ion peak in the mass spectroscopy at 57 m/z indicates that this molecule has the chemical formula $C4H9$, which is that of a stable cation. A bromine atom is characterized by its (M+1) peak at 80. The molecular weight of 2-bromobutane is consistent with the molecular ion peaks at 136 and 138 m/z.

FTIR-

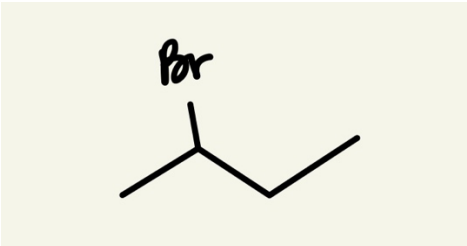
There were just two prominent spikes in the unknown's infrared spectrum. The absence of broad peaks above 3000$cm-1$ suggests that the compound does not contain any alcohol groups. An alkane carbon to hydrogen single bond can be seen in the IR graph, where it appears as a sharp peak from 3300 $cm-1$ to 2800 $cm-1$. There must be at least one C-H bond in the molecule because this is one of the only significant spikes on the IR spectrum. The absence of spikes on the spectrum between 1600 and 1700 $cm-1$ indicates that the molecule does not contain any double or triple bonds. A significant peak appears at 532 $cm-1$, indicating the presence of a C-Br bond within the molecule. The fact that the spectrum shows no additional peaks suggests that the molecule is an alkane and lacks any additional functional groups.

NMR-

The NMR spectra of the unidentified sample A revealed a graph with four signals. The sextet signal at 4.1 ppm integrates, starting from the left, and reveals that there is one hydrogen proton at this carbon. This indicates that the particular hydrogen particle has five neighboring hydrogen atoms and is the only hydrogen atom on its carbon. After that, the integration of the quintet signal at 1.9 ppm reveals the presence of three hydrogen protons at this carbon. This indicates that there are four neighboring hydrogen protons and three hydrogens at this carbon. After that, a doublet signal was detected at 1.7 ppm, and upon integration, it was discovered that this carbon contained three hydrogen protons. This signal is a doublet, which indicates that it has one neighboring hydrogen proton. Ultimately, it was discovered that there are three hydrogen protons at this carbon when the triplet signal at 1.05 ppm was integrated. There are two hydrogen protons next to the signal since it is a triplet. One can conclude that the compound contains one CH, one $CH2$, and two $CH3$ molecules after considering all of the peaks.

CONCLUSION

After examining the three sets of data, the unknown molecule, compound A, can be identified as 2-bromobutane. The graphs indicated that the molecule consists of an alkane chain with four carbon atoms and a bromine atom bound to the second carbon.



REFERENCES

AIST:Spectral Database for Organic Compounds,SDBS. Aist.go.jp. https://sdbs.db.aist.go.jp/sdbs/cgi-bin/direct_frame_top.cgi.

iChemLabs. *ChemDoodle Web Components | Demos > Calculate Molecule Properties*. Chemdoodle.com. https://web.chemdoodle.com/demos/calculate-molecule-properties#customise-template.

"Infrared Spectroscopy."*ACS Reagent Chemicals*, 2017,https://doi.org/10.1021/acsreagents.2008.

Keller, Leonard, et al. Organic Chemistry 2 Lab Manual- Auburn University. 2023 ed., Macmillan Learning Curriculum Solutions, 2023

Libretexts. "How the Mass Spectrometer Works." *Chemistry LibreTexts*, Libretexts, 15 Aug. 2020,https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Supplemental_Modules_(Analytical_Chemistry)/Instrumentation_and_Analysis/Mass_Spectrometry/How_the_Mass_Spectrometer_Works.